EECS820: Advanced Electromagnetics

Multilayer Structures

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May 06, 2025

Question 1: Film on GaAs Substrate (n = 3.55)

Determine the Film Refractive Index and Thickness

To achieve 100% transmission at normal incidence for a film deposited on a GaAs substrate, the film thickness must satisfy the quarter-wavelength condition. This is given by:

$$d = \frac{\lambda_{\text{film}}}{4}$$

where λ_{film} is the wavelength of light in the film. The relationship between the wavelength in the film λ_{film} and the wavelength in vacuum λ_0 is $\lambda_{\text{film}} = \dfrac{\lambda_0}{n_{\text{film}}}$. Thus, the film thickness can be expressed as:

$$d = \frac{\lambda_0}{4n_{\text{film}}}$$

In order to achieve 100% transmission, the refractive index of the film must be chosen such that the impedance of the film matches that of the substrate. This condition is:

$$n_{\text{film}} = \sqrt{n_{\text{substrate}}}$$

where $n_{\text{substrate}} = 3.55$ is the refractive index of GaAs. Substituting the values:

$$n_{\text{film}} = \sqrt{3.55} \approx 1.88$$

Finally, the thickness of the film can be calculated as:

$$d = \frac{980\,\text{nm}}{4 \times 1.88} \approx 130.03\,\text{nm}$$

Thus, the refractive index of the film is $n_{\text{film}} \approx 1.88$ and the film thickness is $d \approx 130.03\,\text{nm}$.

Reflection when the Film Thickness is Half-Wave

For a film thickness equal to half the wavelength in the film, the thickness is given by:

$$d_{\text{half-wave}} = \frac{\lambda_{\text{film}}}{2} = \frac{\lambda_0}{2n_{\text{film}}}$$

Substituting the values for $\lambda_0 = 980\,\text{nm}$ and $n_{\text{film}} \approx 1.88$, we get:

$$d_{\text{half-wave}} = \frac{980\,\text{nm}}{2 \times 1.88} \approx 260.06\,\text{nm}.$$

Next, the reflection at normal incidence is given by the formula:

$$R = \left| \frac{n_{\text{film}} - n_{\text{substrate}}}{n_{\text{film}} + n_{\text{substrate}}} \right|^2$$

where $n_{\text{substrate}} = 3.55$ is the refractive index of GaAs. Substituting the values:

$$R = \left| \frac{1.88 - 3.55}{1.88 + 3.55} \right|^2 \approx 0.094$$

Thus, the reflection for a half-wave film thickness is approximately $R = 0.094$ or 9.4%.

Reflectance Calculation and Plotting for Two Cases

(a) Wavelength Range (900–1060 nm) at Normal Incidence

For normal incidence, the reflection is given by:

$$R(\lambda) = \left| \frac{n_{\text{film}} - n_{\text{substrate}}}{n_{\text{film}} + n_{\text{substrate}}} \right|^2$$

For the wavelength range of 900 - 1060 nm, the reflection is constant and calculated as $R(\lambda) \approx 0.094$.

(b) Angle of Incidence Range (0 - 45°) at 980 nm for TE and TM Polarizations

For varying angles of incidence, the reflection for TE and TM polarizations can be calculated using the Fresnel equations. The transmitted angle θ_t is determined using Snell's law:

$$n_1 \sin(\theta) = n_2 \sin(\theta_t)$$

The reflection coefficients for TE and TM polarizations are:

$$R_{\text{TE}} = \left| \frac{n_1 \cos(\theta) - n_2 \cos(\theta_t)}{n_1 \cos(\theta) + n_2 \cos(\theta_t)} \right|^2$$

$$R_{\text{TM}} = \left| \frac{n_1 \cos(\theta_t) - n_2 \cos(\theta)}{n_1 \cos(\theta_t) + n_2 \cos(\theta)} \right|^2$$

The reflections for TE and TM polarizations are plotted as functions of the angle of incidence from 0 - 45° at 980 nm.

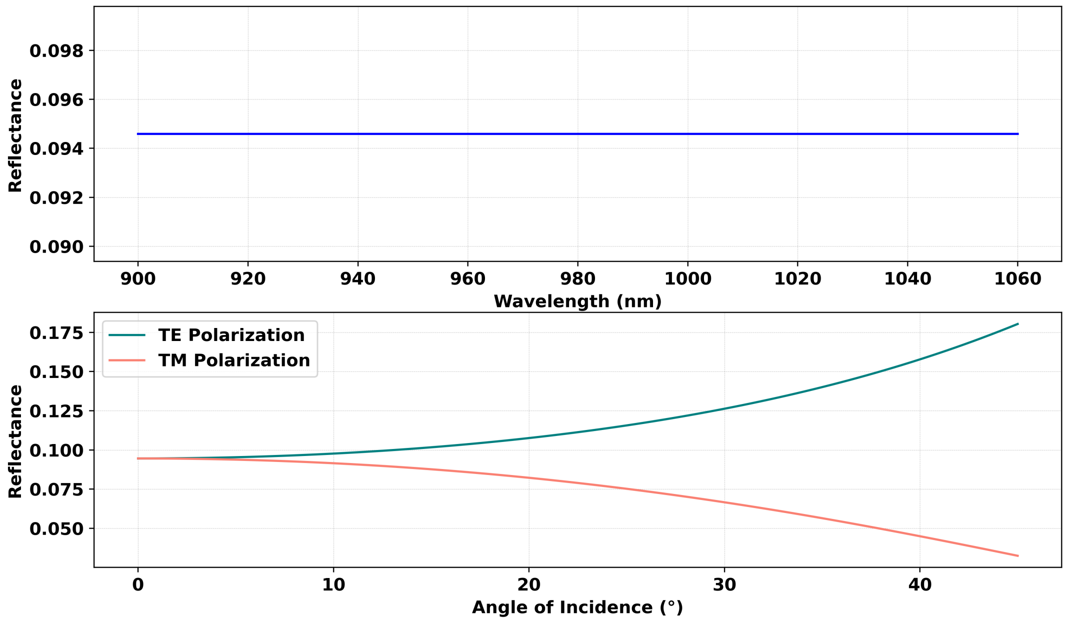

Figure 1: Reflection Coefficient vs Wavelength and Angle of Incidence.

Film Refractive Index and Thickness for Zero Reflection at 980 nm and 20° Angle of Incidence

Here, we are tasked with determining the refractive index and thickness of the film that would produce zero reflection for 980 nm light at a 20° angle of incidence for both TE and TM polarizations. The reflection for both polarizations is given by the Fresnel equations. The transmitted angle θ_t is determined using Snell's law; $n_1 \sin(\theta) = n_2 \sin(\theta_t)$; where n_1 is the refractive index of the film, and $n_2 = 3.55$ is the refractive index of GaAs. For TE and TM polarizations, the reflection coefficients are given in previous section. The goal is to minimize these reflection coefficients for both polarizations at a 20° angle of incidence. To achieve this, an optimization approach was used to find the refractive index n_{film} and thickness d_{film} that minimize the sum of reflection errors for both polarizations. The optimization was performed using the Python 'scipy.optimize' library, with the following steps:

- The objective was to minimize the reflection error for both TE and TM polarizations.
- The optimization started with initial guesses for $n_{\text{film}} = 1.88$ and $d_{\text{film}} = 130.03\,\text{nm}$.
- The refractive index n_{film} was bounded between 1.0 and 3.0, and the thickness d_{film} was bounded between 50 nm and 500 nm.

The result of the optimization process provided the following optimized values:

$$n_{\text{film}} \approx 3.0 \quad \text{and} \quad d_{\text{film}} \approx 130.03\,\text{nm}$$

The optimization minimized the reflection coefficients for both polarizations, resulting in an error of approximately 0.0143, indicating near-zero reflection. Thus, the refractive index and thickness that produce zero reflection at a 20° angle of incidence for both polarizations are $n_{\text{film}} \approx 3.0$ and $d_{\text{film}} \approx 130.03\,\text{nm}$.

Question 2: Laser Mirror Construction

(a) The Thickness of Each of the Two Layers (MgF_3 and SiO_2)

For a quarter-wave multilayer mirror, the thickness of each layer is given by the equation:

$$d_{\text{layer}} = \frac{\lambda_0}{4n_{\text{layer}}}$$

where $\lambda_0 = 1000\,\text{nm}$ is the wavelength in vacuum and n_{layer} is the refractive index of the layer. The thickness of the MgF_3 layer ($n_{\text{MgF3}} = 1.37$) is calculated as:

$$d_{\text{MgF3}} = \frac{1000}{4 \times 1.37} \approx 182.48\,\text{nm}$$

The thickness of the SiO_2 layer ($n_{\text{SiO2}} = 1.46$) is calculated as:

$$d_{\text{SiO2}} = \frac{1000}{4 \times 1.46} \approx 171.23\,\text{nm}$$

Thus, the thicknesses of the layers are $d_{\text{MgF3}} \approx 182.48\,\text{nm}$ and $d_{\text{SiO2}} \approx 171.23\,\text{nm}$.

(b) Minimum Number of Layers to Produce 99.9% Reflected Power

The reflectivity $R(N)$ for a multilayer stack with N pairs of alternating layers is given by the formula:

$$R(N) \approx \left(1 - \left(\frac{n_{\text{layer}}}{n_{\text{substrate}}}\right)^2\right)^{2N}$$

where n_{layer} is the refractive index of the alternating layers (either MgF_3 or SiO_2), $n_{\text{substrate}} = 1.45$ is the refractive index of the glass substrate. The goal is to achieve a reflection of at least 99.9%, i.e., $R(N) \geq 0.999$. To find the minimum number of pairs N, we solve for N such that:

$$(1 - R(N)) = \left(1 - \left(\frac{n_{\text{layer}}}{n_{\text{substrate}}}\right)^2\right)^{2N} = 0.0001$$

Substitute the given values for the refractive indices of MgF_3 ($n_{\text{MgF3}} = 1.37$) and the substrate ($n_{\text{substrate}} = 1.45$), we can solve for N:

$$N = \frac{\ln(0.0001)}{2\ln\left(1 - \left(\frac{n_{\text{MgF3}}}{n_{\text{substrate}}}\right)^2\right)} \approx 2$$

Thus, at least 2 pairs of alternating layers are required to achieve 99.9% reflected power.

(c) Reflected Power for 2, 4, 6, 8, 10, 15, and 20 Pairs of Layers (Alternating MgF_3 and SiO_2)

The reflectivity $R(N)$ for N pairs of alternating layers of MgF_3 and SiO_2 is given by the formula:

$$R(N) = \left(\prod_{i=1}^{N} \frac{1 - \left(\frac{n_{\text{layer}}}{n_{\text{substrate}}}\right)^2}{1 + \left(\frac{n_{\text{layer}}}{n_{\text{substrate}}}\right)^2} \right)^2$$

where n_{layer} alternates between the refractive indices of MgF_3 ($n_{\text{MgF3}} = 1.37$) and SiO_2 ($n_{\text{SiO2}} = 1.46$), and $n_{\text{substrate}} = 1.45$ is the refractive index of the glass substrate. The reflected power for different numbers of alternating layer pairs is as follows:

Number of Layer Pairs (N)	Reflected Power (R)
2	1.52×10^{-7}
4	2.30×10^{-14}
6	3.50×10^{-21}
8	5.31×10^{-28}
10	8.06×10^{-35}
15	5.97×10^{-51}
20	6.50×10^{-69}

As observed, the reflected power decreases rapidly as the number of pairs increases, leading to extremely low reflectivity after a certain point.

Reflected Power for 2, 4, 6, 8, 10, 15, and 20 Pairs of Layers (Alternating MgF_3 and TiO_2)

Here, n_{layer} alternates between the refractive indices of MgF_3 ($n_{\text{MgF3}} = 1.37$) and TiO_2 ($n_{\text{TiO2}} = 2.2$). The reflected power for different numbers of alternating layer pairs is as follows:

Number of Layer Pairs (N)	Reflected Power (R)
2	5.00×10^{-4}
4	2.50×10^{-7}
6	1.25×10^{-10}
8	6.24×10^{-14}
10	3.12×10^{-17}
15	2.50×10^{-26}
20	9.71×10^{-34}

As we can see, the reflectivity decreases rapidly as the number of pairs increases, similar to the previous case with MgF_3 and SiO_2 layers. Changing the order of the layers (i.e., starting with TiO_2 or MgF_3) would result in the same reflectivity because the materials are still alternating and the reflectivity depends on the refractive indices of the individual materials and their interaction with the substrate.

Question 3: Antireflective Coating for a Substrate with Refractive Index $n = 1.5$

The thickness of each layer in the antireflective coating is determined by the quarter-wavelength condition:

$$d_{\text{layer}} = \frac{\lambda_0}{4n_{\text{layer}}}$$

where, $\lambda_0 = 1000\,\text{nm}$ is the wavelength in vacuum, n_{layer} is the refractive index of the layer (either MgF_3 or SiO_2). The thickness of each layer is calculated as follows:

For MgF_3 ($n_{\text{MgF3}} = 1.37$):

$$d_{\text{MgF3}} = \frac{1000}{4 \times 1.37} \approx 182.48\,\text{nm}$$

For SiO_2 ($n_{\text{SiO2}} = 1.46$):

$$d_{\text{SiO2}} = \frac{1000}{4 \times 1.46} \approx 171.23\,\text{nm}$$

The reflectivity $R(N)$ for N pairs of alternating layers of MgF_3 and SiO_2 is given by the formula in previous section. Here, $n_{\text{substrate}} = 1.5$ is the refractive index of the glass substrate. To minimize the reflection, we aim to achieve $R(N) < 0.01$ (i.e., less than 1% reflection). By iterating through increasing values of N and calculating the reflectivity for each case, the minimum number of pairs required to achieve this reflection is $N_{\text{required}} = 1$. For this case, the minimum reflection is:

$$R_{\min} \approx 0.0082 \quad \text{or} \quad 0.82\%$$

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